

Financial Index Analysis of NVIDIA

Volatility Modeling & Risk Analysis

Big Data for Operational Management – Project 2

SEBASTIÃO CLEMENTE

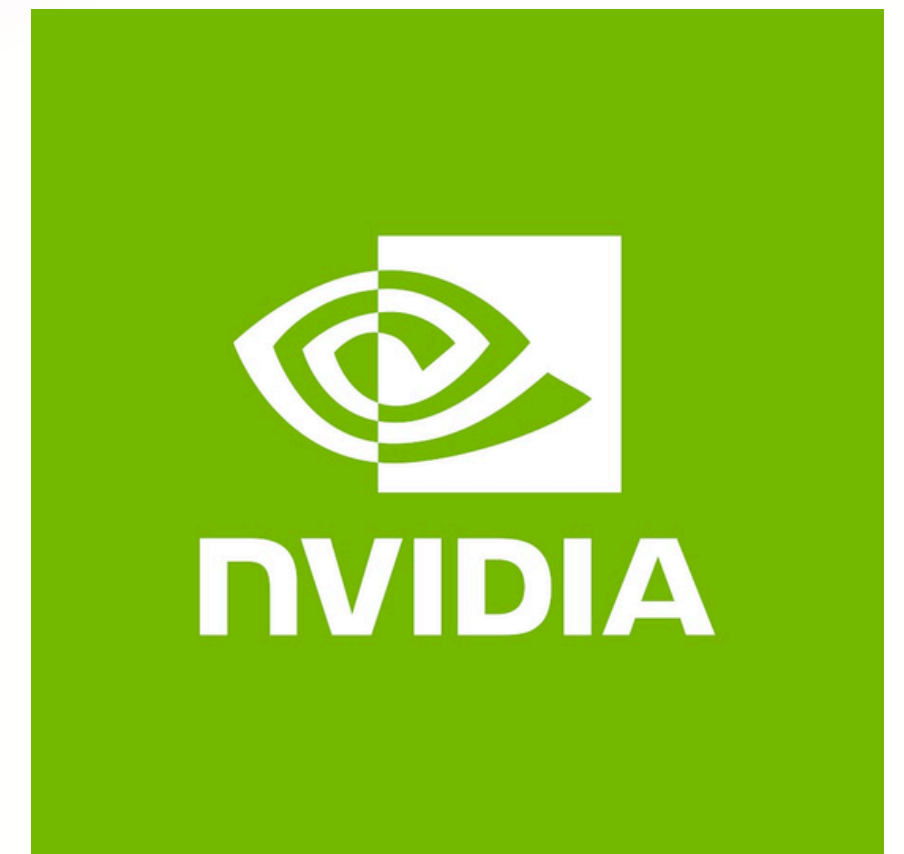
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NVIDIA Dataset

Source Yahoo Finance

Period: 2014 to 2025

Why this dataset?

- *Large price fluctuations*
- *Heavy-tailed returns*
- *Strong volatility clustering*
- *Highly relevant in tech/AI markets → important for risk analysis*



About NVIDIA

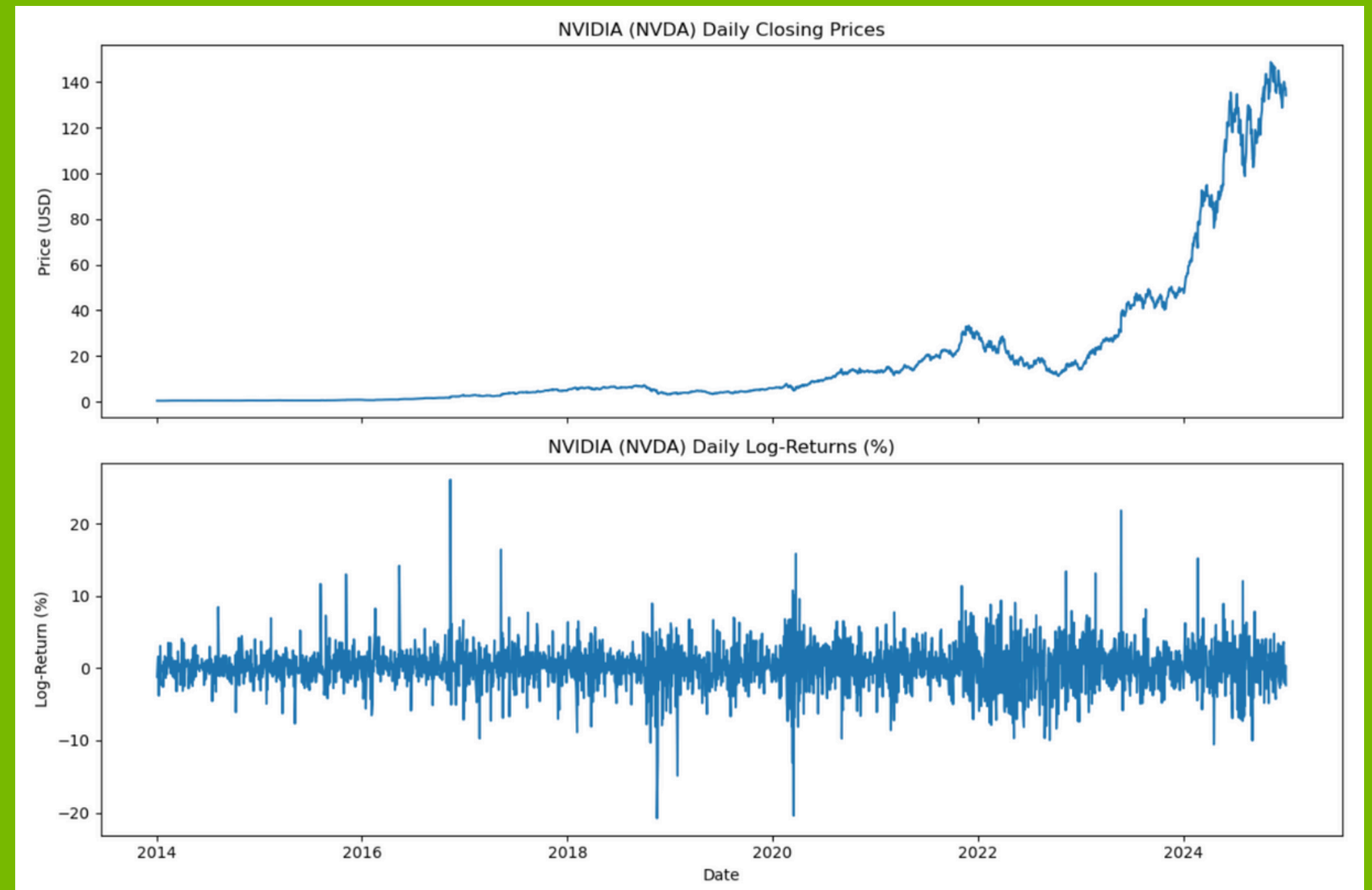
NVIDIA is a global leader in graphics processing units (GPUs), data-center computing, and AI acceleration. The company's rapid growth, driven by surging demand for AI and high-performance computing, has made its stock one of the most influential and volatile in modern equity markets.

Exploratory Data Analysis

Observations

- 1 Price shows rapid long-term growth
- 2 Log-returns show frequent large spikes
- 3
- 4 Absolute returns + rolling std dev reveal volatility clustering

Implication: Volatility is not constant → motivates GARCH



Observations

Model Choice

GARCH

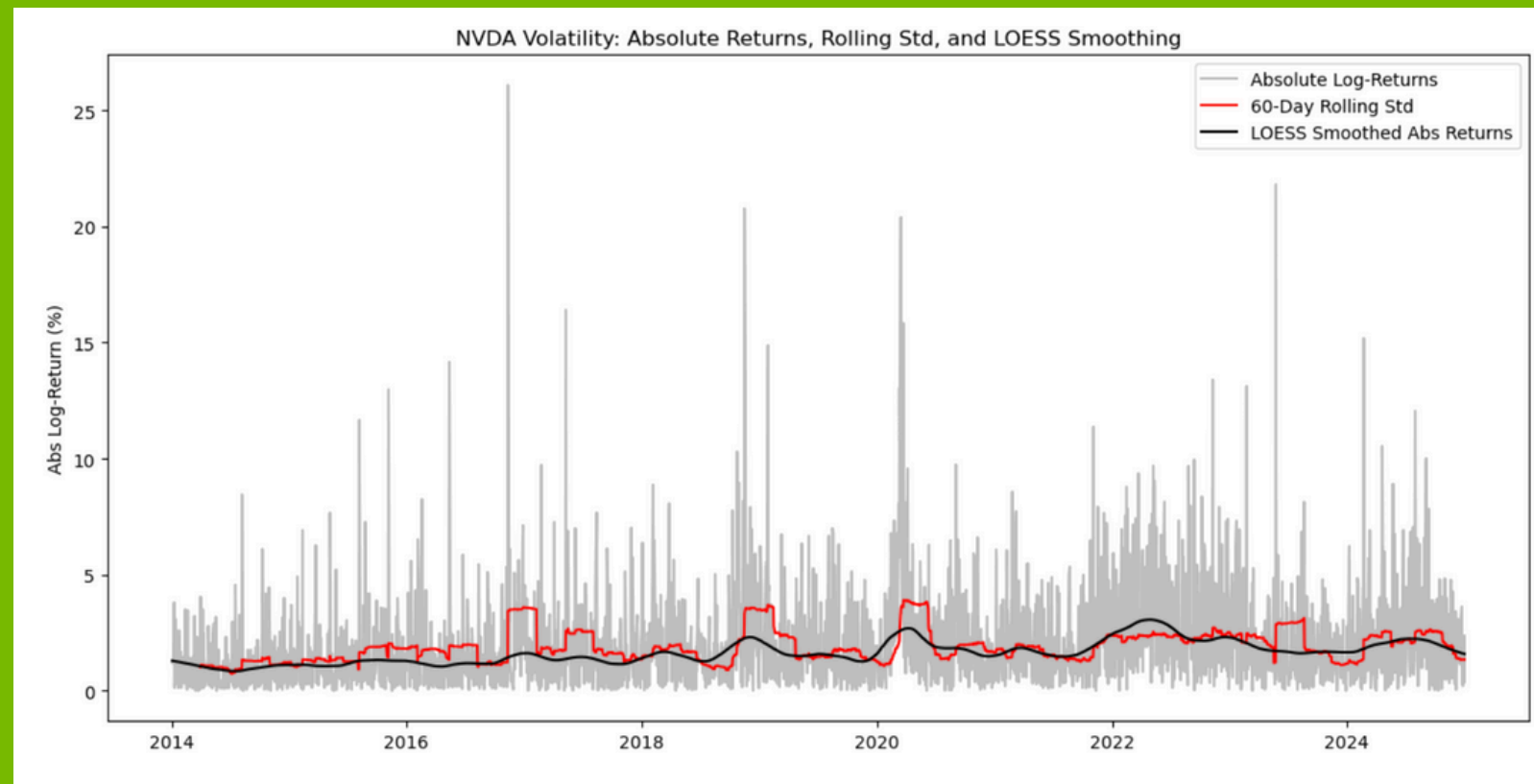
Residuals

VaR

ES & VaR

Forecasting

Why GARCH? (Model Choice)



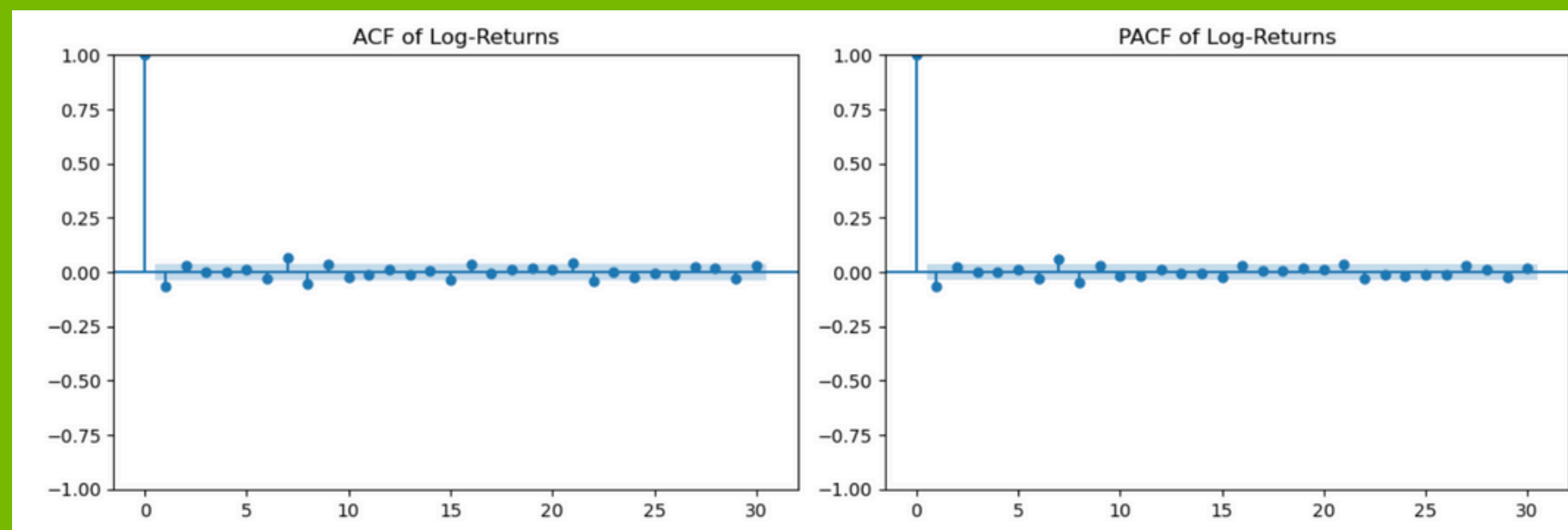
Evidence from EDA

- 1 ACF of squared returns significant
- 2 Variance changes over time
- 3 Heavy-tailed behavior

ARCH vs GARCH

- ARCH(1): captures short-term shocks
- GARCH(1,1): captures persistence, matches financial markets

Chosen model: AR(1) + GARCH(1,1) with t-distributed errors



ARCH(1) Model Summary:

AR - ARCH Model Results			
Dep. Variable:	Log_Return	R-squared:	0.004
Mean Model:	AR	Adj. R-squared:	0.004
Vol Model:	ARCH	Log-Likelihood:	-6833.52
Distribution:	Normal	AIC:	13675.0
Method:	Maximum Likelihood	BIC:	13698.7
		No. Observations:	2766
Date:	Mon, Dec 15 2025	Df Residuals:	2764

Model Summary:

AR - GARCH Model Results			
Dep. Variable:	Log_Return	R-squared:	0.003
Mean Model:	AR	Adj. R-squared:	0.003
Vol Model:	GARCH	Log-Likelihood:	-6502.24
Distribution:	Standardized Student's t	AIC:	13016.5
Method:	Maximum Likelihood	BIC:	13052.0
		No. Observations:	2766
Date:	Mon, Dec 15 2025	Df Residuals:	2764
Time:	10:01:06	Df Model:	2

Observations

Model Choice

GARCH

Residuals

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Forecasting

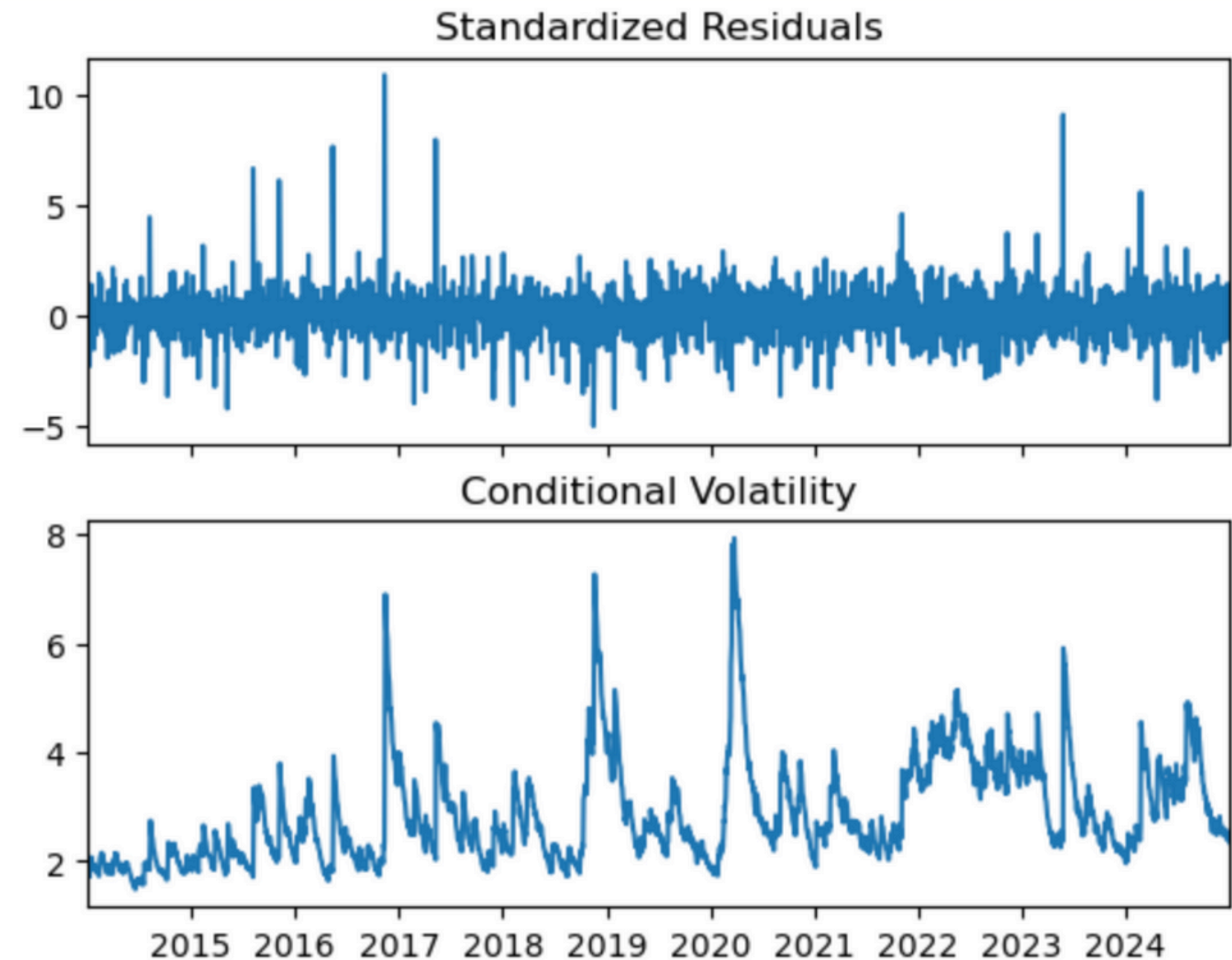
GARCH(1,1) Results

Key Findings:

- α and β significant
- $\alpha + \beta \approx 1 \rightarrow$ volatility is highly persistent
- t-distribution handles heavy tails
- GARCH improves over ARCH based on AIC/BIC

Interpretation:

- Periods of high volatility tend to continue
- Model captures NVIDIA's volatility behavior well



Observations

Model Choice

GARCH

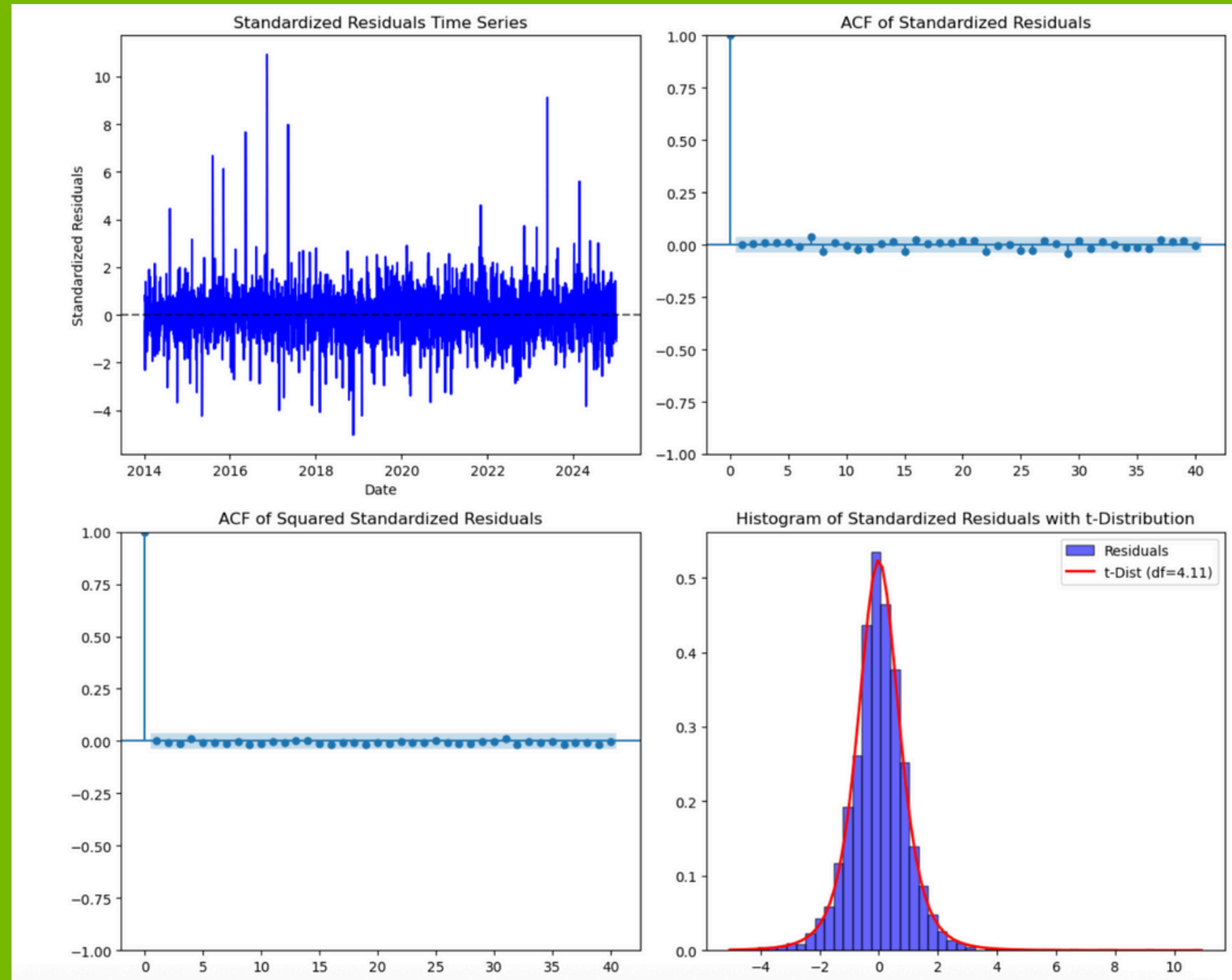
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Forecasting

Analyzing the Residuals



- Standardized residuals fluctuate around zero → no pattern
- ACF of residuals: no significant spikes → mean modeled well
- ACF of squared residuals: no leftover volatility → GARCH captured clustering
- Histogram matches t-distribution → appropriate choice for heavy tails

Observations

Model Choice

GARCH

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Forecasting

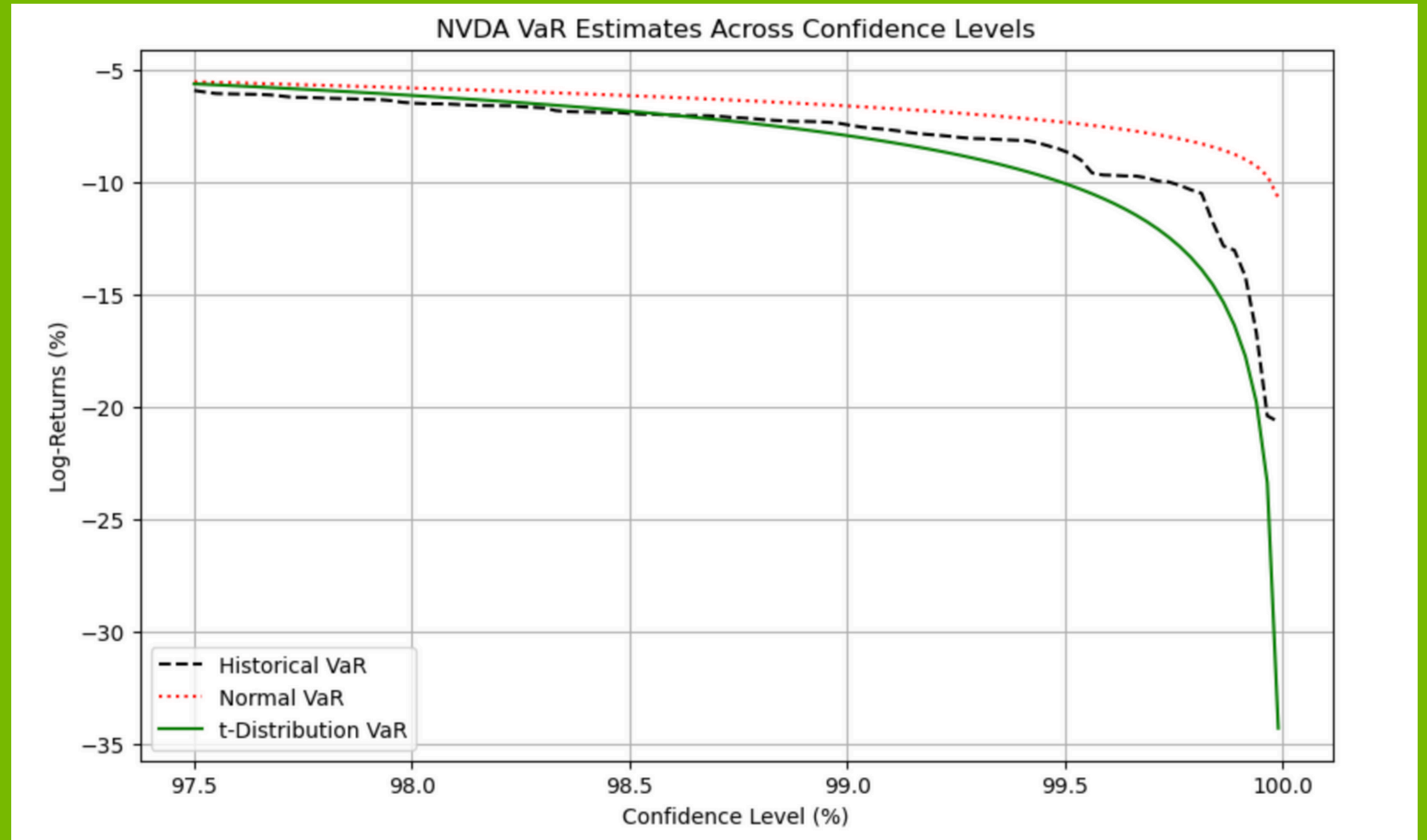
Value at Risk (VaR)

What VaR tells us

- Maximum expected loss at a given confidence level

Results:

- Losses grow rapidly at higher confidence levels
- t-distribution VaR > normal VaR → heavier tails
- NVIDIA can experience large downside movements



Observations

Model Choice

GARCH

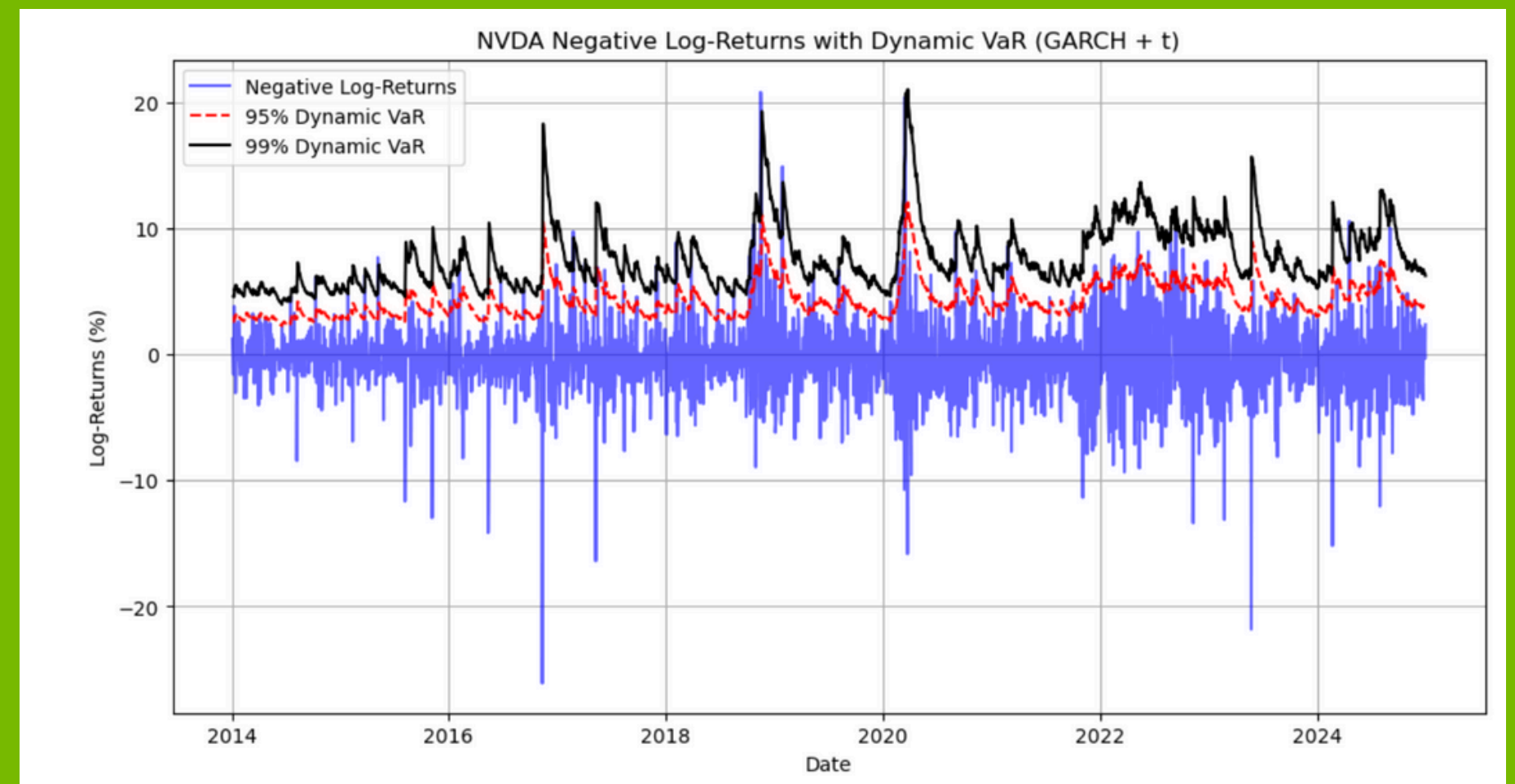
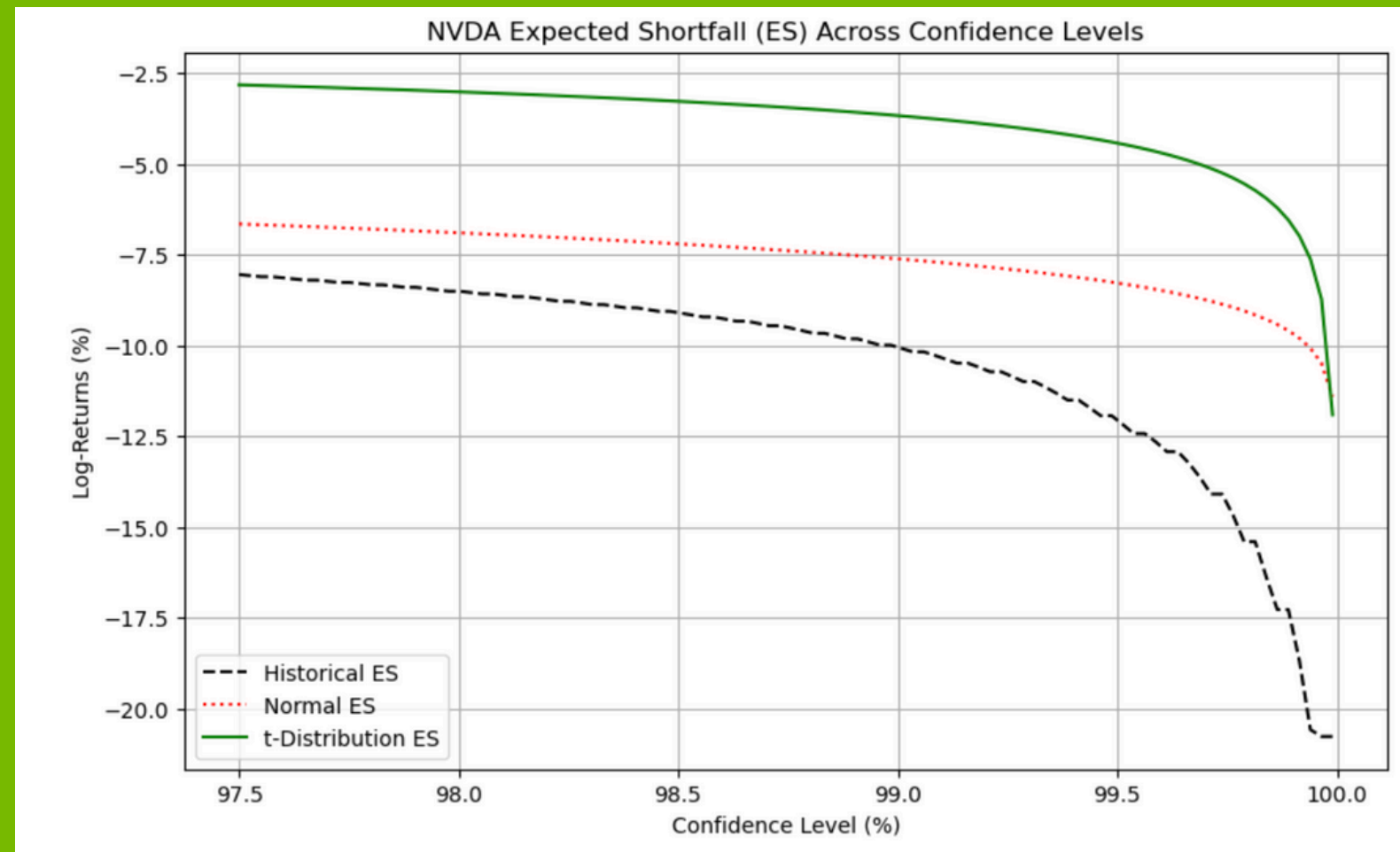
Residuals

VaR

ES & VaR

Forecasting

Expected Shortfall & Dynamic VaR



Expected Shortfall

- Average loss when VaR is exceeded
- Always more negative than VaR → more conservative

Dynamic VaR

- VaR increases during volatile periods
- Shows how risk changes over time
- GARCH tracks real market conditions well

Observations

Model Choice

GARCH

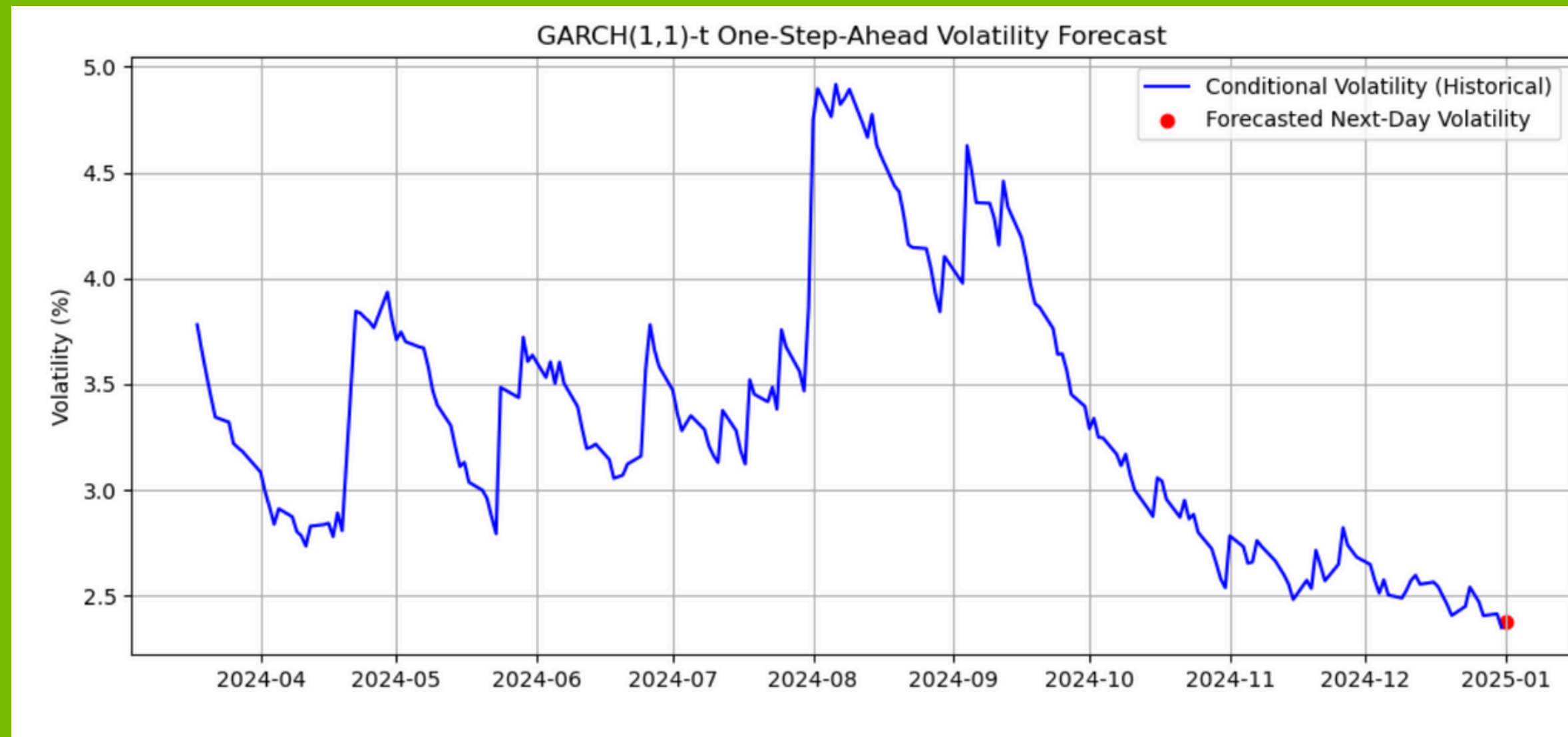
Residuals

VaR

ES & VaR

Forecasting

Forecasting & Model Evaluation



- GARCH(1,1) provides next-day volatility forecasts
- Forecast uses today's information to estimate tomorrow's risk
- NVIDIA's forecasted volatility continues recent volatility patterns
- Useful for short-term risk management and planning
- Supports VaR and ES by providing expected future variance

Next-Day Forecasts from GARCH(1,1)-t Model

Forecasted Mean Return:	0.294305
Forecasted Volatility (σ):	2.376860
Forecasted Variance (σ^2):	5.649465

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Thank You!

Section 1 Group 8 – Project 2

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Appendix

